Photosynthesis

Photosynthesis is the name we give to a special process that green plants use to make food using sunlight, or **light energy**. They use light energy to change **carbon dioxide** and **water** into a special sugar called **glucose**. Plants get their energy from glucose, and this energy allows them to grow and carry out all sorts of functions.

Green plant cells contain a green chemical called **chlorophyll**, which traps light energy from the sun. During photosynthesis, water molecules are split apart using this light energy. The hydrogen parts of each water molecule join with carbon dioxide to make glucose, the special sugar that plants have as their food. The leftover oxygen part of each water molecule is released by the plant through its leaves.

The word photosynthesis is made up of two parts – *photo* means 'light' and *synthesis* means 'to make something'.

☞ **1** Green plants use sunlight to turn what two things into food? The first letter has been given for each.

w _____

c _____

2 Green plants produce what two things as a result of photosynthesis? The first letter has been given for each.

g _____

o _____

3 Which of the things in Question 2 is used by the plant as food?

_____

4 Which of the things in Question 2 is released as a waste product?

_____

5 Write a word equation for what happens in photosynthesis. Use your answers from Questions 1 and 2, plus the ingredient sunlight. Use '+' and '=' symbols.

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Photosynthesis CONT'D

Chlorophyll and sunlight

Sunlight is a form of energy. The sunlight we see is actually made up of different wavelengths of light, which create three different colours – green, red and blue.

The chlorophyll in the cells in a plant's leaves absorb the energy from blue light and red light. They reflect green light back without absorbing it. This is why plants look green to us. They are reflecting light from the green range of the spectrum back at us.

6 Write the colours of light into the correct spaces in the table.

red light blue light green light

Colour of light	Usefulness to plants
	reflected back without being absorbed
	absorbed by the plant's leaves
	absorbed by the plant's leaves

7 Are the following statements true or false?

a) During photosynthesis, water molecules are split apart. True False

b) Plants use light energy from the sun to make carbon dioxide. True False

c) The oxygen part of water is released as a gas. True False

d) The hydrogen parts of water join with carbon dioxide to make glucose. True False

e) Glucose is a mineral from the soil that plants use as their food. True False

f) Glucose is a special sugar made during photosynthesis. True False

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